Vaibhav Satish

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Education

University of California, Irvine: Donald Bren School of Information and Computer Science — Irvine, CA
Bachelor of Science (B.S.) in Computer Science — *Aug 2024 - Expected Mar 2027*

- **GPA:** 3.873/4.0
- **Achievements:** Dean's Honors List (**6/6** consecutive academic quarters)
- **Relevant Coursework:** Data Structures, Algorithms, Advanced Python, C++, Computer Organization, Machine Learning, Intro to Artificial Intelligence, Information Retrieval, Database Management

Experience

Software Engineering / Machine Learning Intern — Fremont, CA
Stealth AI — *Jun 2026 - Present*

- Designed and shipped an **OpenAPI**-based API documentation system that auto-generates endpoint documentation from **JSDoc comments**, integrated into **CI/CD** and adopted company-wide across **12+** endpoints by the entire development team, including the **CTO** and senior engineers.
- Trained and iteratively tuned a **Multilayer Perceptron (MLP)** model outputting **13** body measurements, preprocessing training datasets ranging from **20 to 6,200 rows** and reducing **Mean Absolute Error (MAE)** across most measurement targets through hyperparameter tuning.
- Evaluated per-measurement MAE trade-offs after retraining, identifying which measurements met the accuracy bar for production deployment and flagging others with regressed accuracy for further iteration.
- Extended the measurement pipeline with **8** additional geometrically-calculated length measurements and **4** additional circumference measurements via an **anthropometric prior fusing system**.
- Seeded **NeonDB** and **GCP Cloud SQL** databases across **10+** backend endpoints and shipped an optional **brand-filtering** feature enabling brand-specific queries platform-wide with zero added latency.
- Leveraged the **Cursor AI IDE** and LLM-driven prompt workflows to automate boilerplate code generation and accelerate unit test writing.

Software Developer Research Assistant — Irvine, CA
UC Irvine Charlie Dunlop School of Biological Sciences — *Mar 2026 - Present*

- **Scaled the MedTech@UCI cognitive project into a formal lab partnership**, integrating our cognitive game into the MindCycle app, now used in **30+** active clinical trials.
- Collaborated with the **UCI SLEEP Lab** under **Dr. Katherine Simon**, building backend logging systems to track **4+** core behavioral metrics across **10+** study sessions, improving data-capture reliability.

Software Engineer → Software Development Co-Lead (Promoted Sep 2025) — Irvine, CA
MedTech@UCI — *Sep 2024 - Present*

- Led a team of **5** engineers using **Unity** and **C#** to develop a pediatric cognitive analysis app with image-recognition and memory tools, used by **20+** patients/testers, enabling earlier detection of cognitive delays.
- Designed interactive cognitive games using **React** and **JavaScript**, delivering therapeutic interfaces for dementia and Alzheimer's patients.
- Engineered a **React metrics-tracking engine** to capture **15+** core cognitive metrics, providing clinicians with actionable analytical insights.

Software Development Intern & Instructor — Seattle, WA
Tech Academy of Minnesota — *Jun 2025 - Aug 2025*

- Delivered **full-stack** curriculum to **18** students, guiding web development using **Python**, **HTML**, **CSS**, and **JavaScript**.
- Formulated structured **debugging** frameworks and **12** lab exercises, driving student project completion rates to **100%** for students starting with no prior coding experience.
- Translated complex **software engineering paradigms** into digestible concepts, providing students with a strong foundation of programming fundamentals.

Artificial Intelligence / Machine Learning Researcher & Developer — Remote
Stanford University (via Lumiere Research Program) — *May 2023 - Nov 2023*

- Preprocessed and engineered features from a high-dimensional, multi-modal **NFL injury dataset** (**270,000 rows**) using **Pandas** and **NumPy**, isolating critical risk factors that informed subsequent modeling.
- **Developed**, **trained**, and **tuned Softmax Regression** and **Deep Neural Network** models in **PyTorch** and **Scikit-Learn**, improving predictive accuracy from a **55%** baseline to **69%** (**+14%**).
- Authored technical documentation detailing model limitations and architectural bottlenecks, enabling the team to streamline the next data pipeline and reduce development time.

Artificial Intelligence / Machine Learning Researcher & Developer Remote
Georgia Institute of Technology & Cornell University (via Inspirit AI Research Program) *Jun 2023 - Aug 2023*

- Researched **NLP** and **computer vision** subfields using **Python**, **Scikit-Learn**, and **TensorFlow**, identifying techniques that reduced processing latency for vision tasks.
- Built and validated **3+ computer vision** models in **PyTorch**/**TensorFlow** to detect distracted driving, consistently achieving **80–85%** validation accuracy, and optimized inference pipelines for real-time edge deployment on prototype hardware.
- Defended technical findings and architectural choices to university faculty, incorporating feedback to refine model accuracy.

Projects

ZOTNest - UC Irvine's 2026 IrvineHacks Hackathon Feb 2026
React, Python, FastAPI, Supabase, Gemini Embeddings, RAG, Leaflet

- Built a full-stack property search web app for off-campus housing discovery, targeting **30,000+** UCI students, combining **multi-criteria filtering** with a **Retrieval-Augmented Generation (RAG)** pipeline for semantic search.
- Embedded structured listing data and scraped 100+ Google reviews per property via **Gemini embeddings** (1536-dim), storing vectors in a **Supabase pgvector** store for similarity search.
- Designed a two-stage query pipeline combining **LLM-based** structured preference extraction (budget, bedrooms, distance, shuttle access) with vector similarity search, merging both signals into a normalized hybrid relevance score, achieving **90%** query accuracy.
- Integrated **Leaflet** for real-time property mapping, and processed scraped reviews into a combined per-property document representation to power qualitative "vibe" search.

UCI Themed Geoguessr Dec 2024
React, Node.js, TypeScript, C++, Crow API, Google Cloud Platform, Vercel

- Led a **4-5-person** engineering team from **system architecture ideation** through implementation and deployment, managing structured Git workflows.
- Architected an asynchronous **C++ backend** using **Crow API**, hosted on **Google Cloud Platform**, reducing gameplay latency to sub-**600ms** and supporting **109** real-time concurrent asset updates.
- Constructed a custom mathematical map-scoring algorithm and integrated **Leaflet**, achieving sub-second asset rendering for all users.

Driver Delineation - UC Irvine's 2025 Datathon Apr 2025
Python, PyTorch, TensorFlow, Scikit-Learn, NumPy

- Trained supervised **ML models** using **PyTorch** and **Scikit-Learn** to predict driver risk likelihood from high-dimensional telemetry datasets.
- Executed rigorous **feature engineering** across **40+** impact variables to minimize model overfitting and maximize generalizability.
- Designed a multi-layer **Deep Neural Network (DNN)** in **TensorFlow**, tuning hyper-parameters to achieve **90%** validation accuracy.

Wordle Jun 2026
JavaScript, Python (FastAPI), CSS, React, Next.js

- Deployed a **Wordle** clone with a **FastAPI** backend handling word validation, answer selection, and game-state logic via a **REST API**.
- Built a responsive **React** frontend with modular **JavaScript** and dynamic DOM manipulation, including real-time keyboard-state tracking and color-coded guess feedback.
- Added word-definition lookups via a **Python** backend scraping **nltk.corpus (WordNet)**, plus a customizable word-length feature with a dynamic grid UI in **React**.

Technical Skills

Languages/Frameworks: Java, **Python**, **C++**, SQL, **TypeScript**, **JavaScript**, **React**, Node.js, Next.js, **FastAPI**, **REST APIs**, **OpenAPI**, C#

AI/ML: **Machine Learning**, **Supervised/Unsupervised Learning**, **NLP**, **Computer Vision**, **PyTorch**, **TensorFlow**, **Scikit-Learn**, **Pandas**, **NumPy**, **Data Analysis**

Databases/Cloud Infrastructure: MySQL, PostgreSQL, Supabase, **Google Cloud Platform**, **Vercel**

Other: HTML, CSS, **Docker**, **Git**, Unity, **Agile**, **CI/CD**, Software Engineering Paradigms